5.8 Coordinate Proofs

Geometry 5.8 · Student sheet

Name _____ Period ____ Date _____

I can…

- place a figure in the coordinate plane in a convenient way.
- use the Distance and Midpoint Formulas and slopes in a coordinate proof.
- write a coordinate proof with variable coordinates.

Vocabulary · fill in the missing word

Coordinate proof — a proof that places a figure in the coordinate plane and uses coordinates and _____

Part 1 · Discovery: where should the triangle go?

Draw a right triangle with legs 6 and 8 twice: anywhere, and with the right angle at the origin and the legs on the axes.

Placement	Vertices	Side lengths	Easy or hard?
Anywhere			
At the origin, on the axes	(0, 0), (6, 0), (0, 8)		

Part 2 · Notes and examples (fill in each note, then work the example)

Placing figures: put a vertex at the _____ and sides along the axes when you can.

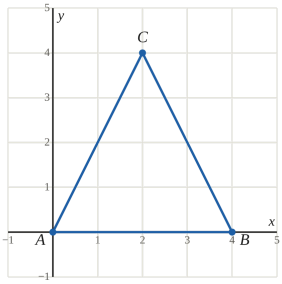
Example 1

Place a 6-by-4 rectangle in the coordinate plane in a convenient way. Give the vertices.

Tools: the Distance Formula shows segments are congruent, the Midpoint Formula finds midpoints, and _____ show segments are parallel or perpendicular.

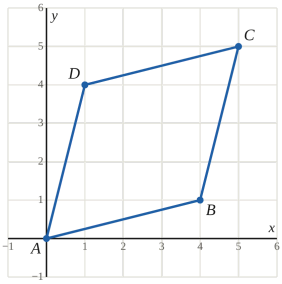
Example 2

Prove that △ABC with A(0, 0), B(4, 0), and C(2, 4) is isosceles.



Example 3

Prove that ABCD with A(0, 0), B(4, 1), C(5, 5), and D(1, 4) is a parallelogram.



Variable coordinates: use letters like (2a, 0) and (0, 2b) to prove a statement for _____ figures of that type.

Example 4

Right $\triangle OAB$ has O(0, 0), A(2a, 0), and B(0, 2b). M is the midpoint of $\overline{AB}$. Prove that M is the same distance from O, A, and B.

Part 3 · Partner practice

Partner 1

Is the triangle with vertices (0, 0), (6, 0), and (3, 5) isosceles? Prove it.

Partner 2

Is the quadrilateral with vertices (0, 0), (3, 0), (4, 2), and (1, 2) a parallelogram? Use slopes.